

## CHAPTER 5

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5.1. Let  $f : A \rightarrow B$  be an integral homomorphism of rings. Show that  $f^* : \text{Spec } B \rightarrow \text{Spec } A$  is a closed mapping. (This is a geometric equivalent of (5.10).)

*Proof.* Take  $\mathfrak{b} \leq B$ , and recall that  $f^*(V(\mathfrak{b})) \subset \overline{f^*(V(\mathfrak{b}))} = V(\mathfrak{b}^c)$  by (Ex. 1.21iii). Now, suppose  $\mathfrak{p} \in V(\mathfrak{b}^c)$  is a prime ideal of  $A$ , i.e.  $\mathfrak{p} \supset \mathfrak{b}^c = f^{-1}(\mathfrak{b})$ . Then in particular  $\mathfrak{p} \supset \ker f$ , so  $\mathfrak{p}$  descends to a prime ideal  $f(\mathfrak{p})$  of  $A/\ker f \cong \text{im } f \subset B$ .

$\text{im } f \subset B$  is integral, so (5.10) gives us a prime  $\mathfrak{q} \in \text{Spec } B$  with  $\mathfrak{q} \cap \text{im } f = f(\mathfrak{p})$ . We have that  $\mathfrak{q} \supset f(\mathfrak{p}) \supset f(\mathfrak{b}^c)$  implies  $\mathfrak{q} \supset \mathfrak{b}^{ce} = \mathfrak{b}$ , i.e.  $\mathfrak{q} \in V(\mathfrak{b})$ . Further,

$$\begin{aligned} f^*(\mathfrak{q}) &= f^{-1}(\mathfrak{q}) \\ &= f^{-1}(\mathfrak{q} \cap \text{im } f) \\ &= f^{-1}(f(\mathfrak{p})) \\ &= \mathfrak{p}, \end{aligned}$$

since  $\mathfrak{p} \supset \ker f$ . Thus,  $\mathfrak{p} \in f^*(V(\mathfrak{b}))$ , so in fact  $f^*(V(\mathfrak{b})) = \overline{f^*(V(\mathfrak{b}))}$  is closed.  $\square$

5.2. Let  $B$  be integral over  $A \subset B$ , and let  $f : A \rightarrow \Omega$  be a homomorphism of  $A$  into an algebraically closed field  $\Omega$ . Show that  $f$  can be extended to a homomorphism of  $B$  into  $\Omega$ .

*Proof.*  $f : A \rightarrow \Omega$  corresponds to a maximal ideal  $\mathfrak{p} = \ker f \leq A$  and an inclusion  $\bar{f} : A/\mathfrak{p} \hookrightarrow \Omega$ . (5.10) supplies us with a prime ideal  $\mathfrak{q} \leq B$  such that  $\mathfrak{q} \cap A = \mathfrak{p}$ , which is in fact maximal by (5.8). Then  $\pi : B \rightarrow B/\mathfrak{q}$  restricts to a map

$$\pi|_A : A \rightarrow \frac{A}{\mathfrak{q} \cap A} = \frac{A}{\mathfrak{p}} \subset \frac{B}{\mathfrak{q}}.$$

(5.6) gives us that  $K := B/\mathfrak{q}$  is integral over  $k := A/\mathfrak{p}$ .

Let  $(k, \bar{f}) \leq (C, \bar{g})$  be a maximal element of the poset  $\Sigma$  of maps from subrings of  $K$  to  $\Sigma$ , ordered by restriction. In particular, it is a valuation ring of  $K$ , so  $K \supset C \supset \bar{k} = K$  by (5.22), whence  $K = C$ . Then  $g := \bar{g} \circ \pi : B \rightarrow \Omega$  has

$$g|_A = \bar{g}|_k \circ \pi|_A = \bar{f} \circ \pi|_A = f.$$

$\square$

5.3. Let  $f : B \rightarrow B'$  be a morphism of  $A$ -algebras, and let  $C$  be an  $A$ -algebra. If  $f$  is integral, prove that  $f \otimes 1 : B \otimes_A C \rightarrow B' \otimes C$  is integral.

*Proof.* We have that  $B'$  is integral over  $f(B) \leq B'$ . That is,  $f(B)[b'_1, \dots, b'_n]$  is finitely generated for all  $b'_1, \dots, b'_n \in B'$ . Note as well that  $(f \otimes 1)(B \otimes C) \cong f(B) \otimes C$ . Then, for arbitrary  $\sum_{i=1}^n b'_i \otimes c_i \in B' \otimes_A C$ ,

$$(f \otimes 1)(B \otimes_A C) \left[ \sum b'_i \otimes c_i \right] \leq f(B)[b'_1, \dots, b'_n] \otimes_A C[c_1, \dots, c_n].$$

The inclusions

$$f(B) \otimes_A C \hookrightarrow f(B)[b'_1, \dots, b'_n] \otimes_A C \hookrightarrow f(B)[b'_1, \dots, b'_n] \otimes_A C[c_1, \dots, c_n]$$

are finite by closure of finiteness under base extension (2.17), so the composition is finite by closure of finiteness under composition (2.16). Then our arbitrary element of  $B' \otimes_A C$  is integral over  $f(B) \otimes_A C$  by (5.1iii), so that  $f \otimes 1$  is integral.  $\square$

5.4 Let  $A \subset B$  be a subring such that  $B$  integral over  $A$ . Let  $\mathfrak{n}$  be a maximal ideal of  $B$  and let  $\mathfrak{m} = \mathfrak{n} \cap A$  be the corresponding maximal ideal of  $A$ . Is  $B_{\mathfrak{n}}$  necessarily integral over  $A_{\mathfrak{m}}$ ?

*Proof.* We will show that this need not hold in general. Consider  $A := k[x^2 - 1] \subset k[x] =: B$ . Clearly  $k[x]$  is integral over  $A$ , as it is a finitely generated  $A$  module. Consider the maximal ideal  $\mathfrak{n} = (x - 1)$ . We have that  $\mathfrak{n} \supset (x^2 - 1)$  and  $\mathfrak{m} := (x^2 - 1)^c$  is maximal in  $A$ , so  $\mathfrak{n} \cap A \supset \mathfrak{m}$  with  $\mathfrak{n}$  a proper ideal implies  $\mathfrak{n} \cap A = \mathfrak{m}$ .

Now, suppose that  $1/(x + 1) \in k[x]_{(x-1)}$  is integral over  $A_{\mathfrak{m}}$ . Then we have  $n \geq 0$  and  $a_0, \dots, a_n \in A_{\mathfrak{m}}$  such that  $(x + 1)^{-n-1} = a_n(x + 1)^{-n} + \dots + a_0$ . Multiplying by  $(x + 1)^{n+1}$ , we get

$$1 = a_n(x + 1) + \dots + a_0(x + 1)^{n+1} \in (x + 1)k[x]_{(x+1)},$$

since  $(x + 1) \cap k[x^2 - 1] = (x^2 - 1)$  by the same argument as above.

Note that  $(x + 1)$  is exactly the kernel of the evaluation map  $f : k[x] \rightarrow k : x \mapsto -1$ . Then  $k[x]_{(x+1)} = k[x]_S$  where  $S = k[x] - \ker f$ , so that  $f(S) \subset k^\times$  gives us  $k_{f(S)} \cong k$ . Hence, we receive a map of localizations

$$k[x]_{(x+1)} \xrightarrow{f} k,$$

under which  $1 \mapsto 1$  and  $a_n(x + 1) + \dots + a_0(x + 1)^{n+1} \mapsto 0$ .  $1 \neq 0$  in  $k$ , so this contradicts our assumption.  $\square$

5.7 Let  $A$  be a subring of  $B$  such that  $B - A$  is closed under multiplication. Show that  $A$  is integrally closed in  $B$ .

*Proof.* Suppose that  $b \in B - A$  is integral over  $A$ , so that  $a_0 = a_1b + \dots + a_nb^n + b^{n+1}$  for  $a_i \in A$  and minimal  $n \geq 0$ . In fact, we may assume  $n > 0$ , as otherwise  $b \in A$ . We have

$$a_0 = b(a_1 + \dots + a_nb^{n-1} + b^n) \in A$$

implies  $a_1 + \dots + a_nb^{n-1} + b^n = a \in A$ , as otherwise  $a_0 \in B - A$  by closure. Then  $a - a_1 = a_2b + \dots + a_nb^{n-1} + b^n$  is of strictly lower degree, contradicting minimality of  $n$ .  $\square$

5.10. A ring homomorphism  $f : A \rightarrow B$  is said to have the *going-up property* (resp. the *going-down property*) if the conclusion of the going-up theorem (resp. going-down theorem) holds for  $B$  and its subring  $f(A)$ . Let  $f^* : \text{Spec } B \rightarrow \text{Spec } A$  be the mapping associated with  $f$ .

(a) Consider the following three statements:

- (i)  $f^*$  is a closed mapping.
- (ii)  $f$  has the going-up property.
- (iii) For any prime ideal  $\mathfrak{q}$  of  $B$ , let  $\mathfrak{p} = \mathfrak{q}^c$ . Then  $f^* : \text{Spec}(B/\mathfrak{q}) \rightarrow \text{Spec}(A/\mathfrak{p})$  is surjective.

Prove that (i)  $\implies$  (ii)  $\iff$  (iii).

(b) Consider the following three statements:

(i')  $f^*$  is an open mapping.

(ii')  $f$  has the going-down property.

(iii') For any prime ideal  $\mathfrak{q}$  of  $B$ , let  $\mathfrak{p} = \mathfrak{q}^c$ . Then  $f^* : \text{Spec}(B_{\mathfrak{q}}) \rightarrow \text{Spec}(A_{\mathfrak{p}})$  is surjective.

Prove that (i')  $\implies$  (ii')  $\iff$  (iii').

*Proof.* (a) We begin by showing that (ii)  $\iff$  (iii). As in the proof of the going-up theorem, it suffices to consider the case of  $\mathfrak{p} \subset \mathfrak{p}' \trianglelefteq A$  and  $\mathfrak{q} \trianglelefteq B$  with  $\mathfrak{q}^c = \mathfrak{p}$ . Applying  $\text{Spec}$  to the following diagram,

$$\begin{array}{ccc} B & \xleftarrow{f} & A \\ \downarrow & & \downarrow \\ B/\mathfrak{q} & \xleftarrow{\bar{f}} & A/\mathfrak{p} \end{array}$$

we receive

$$\begin{array}{ccc} \text{Spec } B & \xrightarrow{f^*} & \text{Spec } A \\ \uparrow & & \uparrow \\ \text{Spec}(B/\mathfrak{q}) & \xrightarrow{f^*} & \text{Spec}(A/\mathfrak{p}). \end{array}$$

Using (1.21iii,iv) and  $f^*(\mathfrak{q}) = \mathfrak{p}$ , we can restrict to

$$\begin{array}{ccc} V(\mathfrak{q}) & \xrightarrow{f^*} & V(\mathfrak{p}) \\ \wr \uparrow & & \wr \uparrow \\ \text{Spec}(B/\mathfrak{q}) & \xrightarrow{f^*} & \text{Spec}(A/\mathfrak{p}). \end{array}$$

Then there exists  $\mathfrak{q}' \supset \mathfrak{q}$  with  $f^{-1}(\mathfrak{q}') = \mathfrak{p}'$  if and only if  $\bar{\mathfrak{p}}' \in \text{im } f^*$ . Hence, surjectivity of  $f^* : \text{Spec}(B/\mathfrak{q}) \rightarrow \text{Spec}(A/\mathfrak{p})$  holds for all prime ideals  $\mathfrak{q} \trianglelefteq B$  if and only if  $f : A \rightarrow B$  possesses the going-up property.

Now, we will show that (i)  $\implies$  (iii). Suppose  $f^*$  is a closed mapping, and recall the last diagram from above. By (1.21iii) once more, we have that

$$\begin{aligned} f^*(V(\mathfrak{q})) &= \overline{f^*(V(\mathfrak{q}))} \\ &= V(\mathfrak{q}^c) \\ &= V(\mathfrak{p}), \end{aligned}$$

giving surjectivity of  $f^* : \text{Spec}(B/\mathfrak{q}) \rightarrow \text{Spec}(A/\mathfrak{p})$ .

(b) We begin by showing that (ii')  $\iff$  (iii'). Once again, we need only consider the case of prime ideals  $\mathfrak{p}' \subset \mathfrak{p} \trianglelefteq A$  with  $\mathfrak{q} \trianglelefteq B$  such that  $\mathfrak{q}^c = \mathfrak{p}$ . Fixing  $\mathfrak{q}$  and  $\mathfrak{p} = \mathfrak{q}^c$ , (3.11iv) gives us the diagram

$$\begin{array}{ccc} \{\mathfrak{q}' \subset \mathfrak{q}\} & \xrightarrow{f^*} & \{\mathfrak{p}' \subset \mathfrak{p}\} \\ \wr \uparrow & & \wr \uparrow \\ \text{Spec } B_{\mathfrak{q}} & \xrightarrow{f^*} & \text{Spec } A_{\mathfrak{p}}, \end{array}$$

whence we conclude our result similarly.

Finally, we show that (i')  $\implies$  (ii'). Suppose  $f^* : \text{Spec } B \rightarrow \text{Spec } A$  is open. Fix  $\mathfrak{p}' \subset \mathfrak{p} \trianglelefteq A$ , and  $\mathfrak{q} \trianglelefteq B$  lying over  $\mathfrak{p}$ , i.e. with  $\mathfrak{q}^c = \mathfrak{p}$ . For any  $b \in B \setminus \mathfrak{q}$ ,  $f^*(X_b) \ni \mathfrak{p}$  is open, so  $\mathfrak{p}' \rightsquigarrow \mathfrak{p}$  implies  $\mathfrak{p}' \in f^*(X_b)$ . This supplies us with  $\mathfrak{q}'_b \in X_b$  such that  $f^*(\mathfrak{q}'_b) = \mathfrak{p}'$ , whence we receive

$$\mathfrak{q}' := \bigcap_{b \in B \setminus \mathfrak{q}} \mathfrak{q}'_b \subset \mathfrak{q}$$

such that

$$f^*(\mathfrak{q}') = \bigcap_{b \in B \setminus \mathfrak{q}} f^*(\mathfrak{q}'_b) = \mathfrak{p}',$$

as desired.  $\square$

5.11 Let  $f : A \rightarrow B$  be a flat homomorphism of rings. Then  $f$  has the going down property.

*Proof.* This is a corollary of (Ex. 3.18) and (Ex. 5.10b).  $\square$

5.16. Let  $k$  be a field and let  $A \neq 0$  be a finitely generated  $k$ -algebra. Then there exist elements  $y_1, \dots, y_r \in A$  which are algebraically independent over  $k$  and such that  $A$  is integral over  $k[y_1, \dots, y_r]$ .

We shall assume that  $k$  is *infinite*. (A different proof is required for the finite case.) From the proof, it follows that  $y_1, \dots, y_r$  may be chosen to be linear combinations of  $x_1, \dots, x_n$ . This has the following geometrical interpretation: if  $k$  is algebraically closed and  $X$  is an affine algebraic variety in  $k^n$  with coordinate ring  $A \neq 0$ , then there exists a linear subspace of dimension  $r$  in  $k^n$  and a linear mapping of  $k^n$  onto  $L$  which maps  $X$  onto  $L$ . [Use Exercise (2).]

5.27 Let  $A, B$  be two local rings.  $B$  is said to *dominate*  $A$  if  $A$  is a subring of  $B$  and the maximal ideal  $\mathfrak{m}$  of  $A$  is contained in the maximal ideal  $\mathfrak{n}$  of  $B$  (or, equivalently, if  $\mathfrak{m} = \mathfrak{n} \cap A$ ).

Let  $K$  be a field and let  $\Sigma$  be the set of all local subrings of  $K$ . If  $\Sigma$  is ordered by the relation of domination, show that  $\Sigma$  has maximal elements, and that  $A \in \Sigma$  is maximal if and only if  $A$  is a valuation ring of  $K$ .

*Proof.* Fix a field  $K$ . We will begin by showing that  $\Sigma$  has maximal elements. By Zorn's lemma, it suffices to show that a chain

$$\begin{array}{ccccccc} A_1 & \leq & A_2 & \leq & \cdots \\ \cup & & \cup & & \\ \mathfrak{m}_1 & \subset & \mathfrak{m}_2 & \subset & \cdots \end{array}$$

must have an upper bound in  $\Sigma$ . We consider the ring  $A := \bigcup A_i$ . Because each  $A_{i+1}$  dominates  $A_i$ , the  $\mathfrak{m}_i$  form an increasing chain of subrings of  $A$ . Their union  $\mathfrak{m} := \bigcup \mathfrak{m}_i$  is an ideal, as  $f \in A_i$ ,  $x \in \mathfrak{m}_j$  implies  $fx \in \mathfrak{m}_{i+j}$ . If  $f \in A \setminus \mathfrak{m}$ , then  $f \in A_i - \mathfrak{m}_i = A_i^\times$  is a unit, whence  $A$  is a local ring with unique maximal ideal  $\mathfrak{m}$  by. Then  $\mathfrak{m} \cap A_i = \mathfrak{m}_i$  gives us  $A_i \leq A$  for all  $i$ , so that  $A \in \Sigma$  is in fact an upper bound to our chain.

Suppose that  $A < B \in \Sigma$  have ideals  $\mathfrak{m} \trianglelefteq A$  and  $\mathfrak{n} \trianglelefteq B$ . It suffices to show that  $A$  is not a valuation ring. Take  $f \in B \setminus A$  supplied by  $B \neq A$ . We have  $f^{-1} \notin A$ , as otherwise  $f^{-1} \in \mathfrak{m} \setminus \mathfrak{n}$  contradicts  $A \leq B$ . Thus,  $A$  is not a valuation ring.

Let  $A \in \Sigma$  be maximal, and suppose it is not a valuation ring. We can take  $x \in K$  such that  $x, x^{-1} \notin A$ . Then  $(\mathfrak{m}, x) \trianglelefteq A[x]$  is a proper ideal of  $A \subsetneq A[x]$ , and is thus contained in

some maximal ideal  $\mathfrak{n} \trianglelefteq A[x]$ . Localizing at  $\mathfrak{n}$ , we receive a local ring  $A' := A[x]_{\mathfrak{n}} \supset A$  with unique maximal ideal  $\mathfrak{n}' := \mathfrak{n}A' \supset \mathfrak{m}$ . Hence,  $A'$  dominates  $A$ , contradicting maximality of  $A$ . We conclude that  $A$  is indeed a valuation ring.  $\square$

5.28 Let  $A$  be an integral domain,  $K$  its field of fractions. Show that the following are equivalent:

- (a)  $A$  is a valuation ring of  $K$ ;
- (b) If  $\mathfrak{a}, \mathfrak{b}$  are any two ideals of  $A$ , then either  $\mathfrak{a} \subset \mathfrak{b}$  or  $\mathfrak{b} \subset \mathfrak{a}$ .

Deduce that if  $A$  is a valuation ring and  $\mathfrak{p}$  is a prime ideal of  $A$ , then  $A_{\mathfrak{p}}$  and  $A/\mathfrak{p}$  are valuation rings of their fields of fractions.

*Proof.* ((a)  $\implies$  (b)) Let  $\mathfrak{a}, \mathfrak{b} \trianglelefteq A$ , and suppose without loss of generality that  $\mathfrak{a} \not\subset \mathfrak{b}$ . Then fix  $f \in \mathfrak{a} \setminus \mathfrak{b}$  and let  $0 \neq g \in \mathfrak{b}$ .  $f/g \notin A$ , so  $A$  a valuation ring gives  $g/f \in A$ , whence  $g \in \mathfrak{a}$ , so that  $\mathfrak{b} \subset \mathfrak{a}$ .

((b)  $\implies$  (a)) Let  $x/y \in K$ , and consider  $(x), (y) \trianglelefteq A$ . If  $(x) \subset (y)$ , then  $x \in (y)$  implies  $x/y \in A$ . If  $(y) \subset (x)$ , then  $y \in (x)$  implies  $y/x \in A$ .

Now, suppose  $A$  is a valuation ring and  $\mathfrak{p} \trianglelefteq A$  is a prime ideal. Any  $\mathfrak{a}, \mathfrak{b} \trianglelefteq A_{\mathfrak{p}}$  are extended, and  $\mathfrak{a}^c \subset \mathfrak{b}^c$  or  $\mathfrak{a}^c \supset \mathfrak{b}^c$  by the above, whence  $\mathfrak{a} \subset \mathfrak{b}$  or  $\mathfrak{a} \supset \mathfrak{b}$ . The ideal lattice of  $A/\mathfrak{p}$  is isomorphic to the poset  $\{\mathfrak{a} \supset \mathfrak{p}\}$ , so it is clear that the second criterion holds.  $\square$

5.29 Let  $A$  be a valuation ring of a field  $K$ . Show that every subring of  $K$  which contains  $A$  is a localization of  $A$ .

*Proof.* Let  $B$  be a subring of  $K$  containing  $A$ . By (5.18i,ii),  $B$  is a valuation ring, and hence local with maximal ideal  $\mathfrak{m} \trianglelefteq B$ . Let  $\mathfrak{n} := A \cap \mathfrak{m}$  be the contraction of  $\mathfrak{m}$  by  $A \hookrightarrow B$ , and consider the localization  $A_{\mathfrak{n}} \subset B$ . If  $f \in B \setminus A$ , then  $f^{-1} \in A$ , so that  $f \in B^{\times} = B \setminus \mathfrak{m}$  implies  $f = (f^{-1})^{-1} \in A_{\mathfrak{n}}$ , whence  $B \subset A_{\mathfrak{n}}$ .  $\square$

5.30 Let  $A$  be a valuation ring of a field  $K$ . The group  $A^{\times}$  of units of  $A$  is a subgroup of the multiplicative group  $K^{\times}$  of  $K$ .

Let  $\Gamma = K^{\times}/A^{\times}$ . If  $\xi, \eta \in \Gamma$  are represented by  $x, y \in K$ , define  $\xi \geq \eta$  to mean  $xy^{-1} \in A$ . Show that this defines a total ordering on  $\Gamma$  which is compatible with the group structure (i.e.,  $\xi \geq \eta \implies \xi\omega \geq \eta\omega$  for all  $\omega \in \Gamma$ ). In other words,  $\Gamma$  is a totally ordered abelian group. It is called the *value group* of  $A$ .

Let  $v : K^{\times} \rightarrow \Gamma$  be the canonical homomorphism. Show that  $v(x + y) \geq \min(v(x), v(y))$  for all  $x, y \in K^{\times}$ .

*Proof.* Clearly  $\geq$  is reflexive and transitive. If  $\xi \geq \eta$  and  $\eta \geq \xi$ , then  $xy^{-1} \in A$  and  $yx^{-1} \in A$  implies  $xy^{-1} \in A^{\times}$ , so that  $\xi = \eta$  gives us anti-symmetry. It is total by definition of a valuation ring. Now, we show it is compatible with the group structure. Let  $\xi \geq \eta$ ,  $\omega \in \Gamma$ , and take representatives  $x, y, z \in K$ . Then  $(xz)(yz)^{-1} = xy^{-1} \in A$  implies  $\xi\omega \geq \eta\omega$ .

Suppose, without loss of generality, that  $v(x) \geq v(y)$ , i.e.  $xy^{-1} \in A$ . Then

$$(x + y)y^{-1} = xy^{-1} + 1 \in A$$

implies  $v(x + y) \geq v(y)$ , as desired. Then  $d(x, y) := -v(x - y)$  defines an ultrametric on  $K$ , thus giving  $K$  the structure of a non-archimedean field.  $\square$

5.31 Conversely, let  $\Gamma$  be a totally ordered abelian group (written *additively*), and let  $K$  be a field. A *valuation of  $K$  with values in  $\Gamma$*  is a mapping  $v : K^{\times} \rightarrow \Gamma$  such that

- (a)  $v(xy) = v(x) + v(y)$ ,
- (b)  $v(x + y) \geq \min(v(x), v(y))$ ,

for all  $x, y \in K^\times$ . Show that the set of elements  $x \in K$  such that  $x = 0$  or  $v(x) \geq 0$  is a valuation ring of  $K$ . This ring is called the valuation ring of  $v$ , and the subgroup  $v(K^\times)$  is the *value group* of  $v$ .

Thus, the concepts of valuation ring and valuation are essentially equivalent.

*Proof.* Define  $A := \{x \in K : x = 0 \text{ or } v(x) \geq 0\}$ . Then  $0 \in A$ ,  $A$  is additive by (b), and multiplicative by (a). If  $x \in K^\times$ , then

$$v(x) + v(x^{-1}) = v(1) = 0$$

implies either  $v(x) \geq 0$  or  $v(x^{-1}) \geq 0$ . That is,  $x \in A$  or  $x^{-1} \in A$ . We conclude that  $A$  is indeed a valuation ring of  $K$ .  $\square$

5.32 Let  $\Gamma$  be a totally ordered abelian group. A subgroup  $\Delta \leq \Gamma$  is *isolated* in  $\Gamma$  if, whenever  $0 \leq \beta \leq \alpha$  and  $\alpha \in \Delta$ , we have  $\beta \in \Delta$ . Let  $A$  be a valuation ring of a field  $K$ , with value group  $\Gamma$ . If  $\mathfrak{p}$  is a prime ideal of  $A$ , show that  $v(A \setminus \mathfrak{p})$  is the set of elements  $\geq 0$  in an isolated subgroup  $\Delta \leq \Gamma$ , and that the mapping so defined of  $\text{Spec } A$  into isolated subgroups of  $\Gamma$  is surjective.

If  $\mathfrak{p} \leq A$  is prime, what are the value groups of the valuation rings  $A/\mathfrak{p}$ ,  $A_{\mathfrak{p}}$ ?

*Proof.* Suppose  $\alpha = v(a) \in v(A \setminus \mathfrak{p})$ , and let  $0 \leq \beta \leq \alpha$  for  $\beta = v(b) \in v(A)$ . Then  $ab^{-1} \in A$ , so  $b \in A \setminus \mathfrak{p}$ , as otherwise  $a = (ab^{-1})b \in \mathfrak{p}$ . Hence,  $\beta \in v(A \setminus \mathfrak{p})$ .

Now, let  $\Delta \leq \Gamma$  be an isolated subgroup, and consider  $\mathfrak{p} := v^{-1}(\Gamma \setminus \Delta) \cup \{0\}$ . Additivity is clear. If  $b \in v^{-1}(\Gamma \setminus \Delta)$  and  $a \in A \setminus \{0\}$ , then  $v(ab) = v(a) + v(b) \geq v(b) \geq 0$ , so  $v(ab) \notin \Delta$  gives us that  $\mathfrak{p}$  is an ideal. Finally, if  $a, b \in A \setminus \mathfrak{p}$ , then  $v(ab) = v(a) + v(b) \in \Delta$  implies  $ab \in A \setminus \mathfrak{p}$ , whence  $\mathfrak{p}$  is prime. It is clear that  $\mathfrak{p} \mapsto \Gamma$  under the mapping of  $\text{Spec } A$ .

The value group of  $A_{\mathfrak{p}}$  is

$$\begin{aligned} \frac{K^\times}{A_{\mathfrak{p}}^\times} &= \frac{K^\times}{A_{\mathfrak{p}} \setminus \mathfrak{p}A_{\mathfrak{p}}} \\ &= \frac{K^\times/A^\times}{(A_{\mathfrak{p}} \setminus \mathfrak{p}A_{\mathfrak{p}})/A^\times} \\ &= \frac{\Gamma}{\pm v(A \setminus \mathfrak{p})} \\ &= \Gamma/\Delta. \end{aligned}$$

Dually, the value group of  $A/\mathfrak{p}$  is

$$\frac{\text{Frac}(A/\mathfrak{p})}{(A/\mathfrak{p})^\times} =$$

$\square$

5.33 Let  $\Gamma$  be a totally ordered abelian group,  $k$  any field, and  $A := k[\Gamma]$  the group algebra of  $\Gamma$  over  $k$ . Show that  $A$  is an integral domain.

If  $u = \lambda_1\alpha_1 + \cdots + \lambda_n\alpha_n$  is any nonzero element of  $A$ , where each  $\lambda_i \neq 0$  and  $\alpha_1 < \cdots < \alpha_n$ , define  $v_0(u) := \alpha_1$ .<sup>1</sup> Show that  $v_0 : A \setminus \{0\} \rightarrow \Gamma$  satisfies conditions (a) and (b) of (Ex. 5.31).

Let  $K$  be the fraction field of  $A$ . Show that  $v_0$  can be uniquely extended to a valuation  $v$  of  $K$ , and that the value group of  $v$  is precisely  $\Gamma$ .

<sup>1</sup>This arises from the natural  $\Gamma$ -graded structure of  $k[\Gamma]$ .

*Proof.* Take  $u, u' \in A$  nonzero and write

$$\begin{aligned} u &= \lambda_1 x_{\alpha_1} + \cdots + \lambda_m x_{\alpha_m}, \\ u' &= \lambda'_1 x_{\alpha'_1} + \cdots + \lambda'_n x_{\alpha'_n} \end{aligned}$$

for  $\lambda_i, \lambda'_i \neq 0$  and  $\alpha_1 < \cdots < \alpha_m, \alpha'_1 < \cdots < \alpha'_n$ . Suppose by way of contradiction that

$$\begin{aligned} 0 &= uu' \\ &= \sum_{i=1}^m \sum_{j=1}^n \lambda_i \lambda'_j x_{\alpha_i + \alpha'_j}. \end{aligned}$$

Then  $\alpha_1 + \alpha'_1 < \alpha_i + \alpha'_j$  for all  $i, j \neq 1$ , so  $\lambda_1 \lambda'_1 x_{\alpha_1 + \alpha'_1} = 0$  implies  $\lambda_1 = 0$  or  $\lambda'_1 = 0$ , a contradiction.

Let  $u, u' \in A$  nonzero. By the above,  $uu' = \lambda_1 \lambda'_1 x_{\alpha_1 + \alpha'_1} + \cdots$ , so  $v_0(uu') = \alpha_1 + \alpha'_1$  gives us (a). For (b), we allow  $m, n = 0$  and induct on  $k = \#\{i \mid \lambda_i + \lambda'_i = 0, \alpha_i = \alpha'_i\}$ . If  $k = 0$ , then clearly  $v_0(u + u') = \min\{v_0(u), v_0(u')\}$ . If  $k = m = n$ , then  $u + u' = 0$ , so the result holds trivially. If only one of  $m, n = 0$ , then without loss of generality  $n = 0$ , so  $v_0(u + u') = v_0(u) = \min(v_0(u), \infty)$ . Otherwise,  $v_0(u + u') = v_0(u'' + u''')$  for  $u'' = u - \lambda_1 x_{\alpha_1}$ ,  $u''' = u' - \lambda'_1 x_{\alpha'_1}$ , and  $k' = k - 1$ , so

$$\begin{aligned} v_0(u + u') &= v_0(u'' + u''') \\ &\geq \min(v_0(u''), v_0(u''')) \\ &\geq \min(v_0(u), v_0(u')), \end{aligned}$$

whence the result follows by induction.

Take  $K = \text{Frac } A$ . Then for any  $x, y \in A$  and  $y \neq 0$ ,

$$v(xy^{-1}) = v((xy^{-1})y) = v(xy^{-1}) + v(y)$$

forces  $v(xy^{-1}) = v_0(x) - v_0(y)$ . Clearly this is a valuation on  $K$ , and  $v : K^\times \rightarrow \Gamma$  is surjective by construction, so  $\Gamma$  is the valuation group of  $v$ .  $\square$

5.34 Let  $A$  be a valuation ring and  $K$  its field of fractions. Let  $f : A \rightarrow B$  be a ring homomorphism such that  $f^* : \text{Spec } B \rightarrow \text{Spec } A$  is a closed mapping. Then if  $g : B \rightarrow K$  is any  $A$ -algebra homomorphism (i.e.,  $g \circ f$  is the embedding  $A \hookrightarrow K$ ), we have  $g(B) = A$ .

*Proof.* Set  $C := g(B)$ .  $g$  an  $A$ -algebra homomorphism gives us  $A \subset C \subset K$ , whence  $C$  is a valuation ring by (5.18ii), and both  $A, C$  are local by (5.18i). Let  $\mathfrak{m} \trianglelefteq A$  and  $\mathfrak{n} \trianglelefteq C$  be their respective maximal ideals.  $f$  is closed by hypothesis, while  $g : B \rightarrow C$  is surjective, and thus a closed embedding. Then the embedding  $A \hookrightarrow C$  is closed, so that  $\mathfrak{n}^c$  is maximal. We conclude that  $\mathfrak{n}^c = \mathfrak{m}$ , so that  $C$  dominates  $A$ , meaning  $C = A$  by maximality of valuation rings under domination (Ex. 5.27). Alternatively, (Ex. 5.29) gives us  $C = A_{\mathfrak{p}}$ , but  $\mathfrak{p} = \mathfrak{p}A_{\mathfrak{p}}^c = \mathfrak{n}^c = \mathfrak{m}$  implies  $C = A_{\mathfrak{m}} = A$ .  $\square$

5.35 From (Ex. 5.1) and (Ex. 5.3), it follows that, if  $f : A \rightarrow B$  is integral and  $C$  is any  $A$ -algebra, then the mapping  $(f \otimes 1)^* : \text{Spec}(B \otimes_A C) \rightarrow \text{Spec } C$  is a closed map.

Conversely, suppose that  $f : A \rightarrow B$  has this property and that  $B$  is an integral domain. Then  $f$  is integral.

Show that the result just proved remains valid if  $B$  is a ring with only finitely many minimal prime ideals (e.g., if  $B$  is Noetherian).

*Proof.* Recall that  $f : A \rightarrow B$  is said to be integral if  $f(A) \subset B$  is integral. Let  $\pi : A \rightarrow A/\ker f$  be the projection and  $\bar{f} : A/\ker f \hookrightarrow B$  the map with  $f = \bar{f} \circ \pi$ . Right exactness of tensor products gives us that  $\pi \otimes 1 : C \rightarrow (A/\ker f) \otimes_A C$  is surjective. Then  $(\pi \otimes 1)^*$  is a closed embedding, so

$$(f \otimes 1)^* = (\pi \otimes 1)^* \circ (\bar{f} \otimes 1)$$

implies

$$\begin{aligned} (\bar{f} \otimes 1)^*(U) &= (\pi \otimes 1)^{*-1}((\pi \otimes 1)^* \circ (\bar{f} \otimes 1)) \\ &= (\pi \otimes 1)^{*-1}((f \otimes 1)^*(U)), \end{aligned}$$

which is closed whenever  $U$  is closed and  $f$  satisfies our property. Thus,  $\bar{f} : A/\ker f \hookrightarrow B$  satisfies our property, so we may reduce to the case of  $A \cong f(A) \subset B$ .

Suppose first that  $A \subset B$  are integral domains,  $B$  has fraction field  $K$ , and that the inclusion  $f : A \hookrightarrow B$  has the above property. We wish to show that the integral closure of  $A$  in  $B$  is precisely  $B$ . It suffices to show that the integral closure  $\bar{A}$  contains  $B$ . By (5.22), it suffices to show that  $B \subset C$  for all valuation rings  $A \subset C \subset K$ .

Take  $A \subset C \subset K$  to be an arbitrary valuation ring of  $K$  containing  $A$ . Consider the maps  $f \otimes 1 : C \rightarrow B \otimes_A C$  and  $g : B \otimes_A C \rightarrow K : b \otimes c \mapsto bc$ . We have that  $g \circ (f \otimes 1) : C \rightarrow K$  is the embedding, and  $(f \otimes 1)^* : \text{Spec } B \otimes_A C \rightarrow \text{Spec } C$  is closed. Thus, (Ex. 5.34) gives us  $g(B \otimes_A C) = C$ , but  $B \subset g(B \otimes_A C)$ , so  $B \subset C$  as desired.

Now, take  $B$  with finitely many minimal prime ideals  $\mathfrak{p}_1, \dots, \mathfrak{p}_n \trianglelefteq B$  and subring  $A \subset B$  such that  $f : A \hookrightarrow B$  has the above property. Then each  $A \rightarrow B \rightarrow B/\mathfrak{p}_i$  has the above property since each  $\pi_i \otimes 1 : B \otimes_A C \rightarrow (B/\mathfrak{p}_i) \otimes_A C$  is surjective and tensor products are functorial. Thus, the  $A \rightarrow B/\mathfrak{p}_i$  are integral, whence  $A \rightarrow \prod B/\mathfrak{p}_i$  is integral by (Ex. 5.6).

The map  $\prod B/\mathfrak{p}_i \rightarrow B/\sqrt{0}$  given by  $(x_i + \mathfrak{p}_i)_{i=1}^n \mapsto \sum_i x_i + \sqrt{0}$  is surjective, so  $A \rightarrow B/\sqrt{0} := \bar{B}$  is integral. It suffices, then, to prove the following lemma.

**Lemma 1.** *If  $\mathfrak{b} \trianglelefteq B$  is an ideal such that  $b \in \mathfrak{b}$  implies  $b^n \in A$  for some  $n > 0$ , and  $A/\mathfrak{b}^c \rightarrow B/\mathfrak{b}$  is integral, then  $A \rightarrow B$  is integral. In particular, if  $A \rightarrow B/\sqrt{0}$  is integral, then  $A \rightarrow B$  is integral.*

*Proof.* Define  $\bar{A} := A/\mathfrak{b}^c$  and  $\bar{B} := B/\mathfrak{b}$ . Then, for each  $x \in B$ , we have a monic polynomial  $\bar{f}(t) \in \bar{A}[t]$  such that  $\bar{f}(\bar{x}) = \bar{0}$ . Take  $f(t) \in A[t]$  which projects to  $\bar{f}$ . Then  $f(x) \in \mathfrak{b}$ , so there exists  $n > 0$  such that  $f(x)^n \in A$ .  $f(t)$  is monic of positive degree, so  $f(t)^n \in A[t]$  is monic of positive degree, whence  $f(t)^n - f(x)^n \in A[t]$  is as well. Clearly  $x$  is a root of  $f(t)^n - f(x)^n$ , so that  $x$  is integral over  $A$ , as desired.  $\square$

We conclude that  $A \rightarrow B$  is integral, completing our proof.  $\square$